coursera | IBM

Develop Generative AI Applications: Get Started ✕

Today's Skill Points 97 XP
See skill progress

Module 1
Foundations of Generative AI and Prompt Engineering

Module 2
Introduction to LangChain in GenAI Applications

LangChain Core Components and Advanced Features

- ✓ Recap: Introduction to LangChain
  Reading • 15 min
- ✓ LangChain: Core Concepts
  Video • 6 min
- ✓ LangChain Chains and Agents for Building Applications
  Video • 6 min
- ✓ (Optional) LangChain LCEL Chaining Method
  Video • 4 min
- ✓ Lab: Build Smarter AI Apps: Empower LLMs with LangChain
  Ungraded App Item • 1h
- ✓ Practice Quiz: LangChain Core Components and Advanced Features
  Practice Assignment • Grade: 100%

Module Summary and Evaluation

- ✓ Summary and Highlights: Introduction to LangChain in GenAI Applications
  Reading • 2 min
- ✓ Cheat Sheet: Introduction to LangChain in GenAI Applications
  Reading • 30 min
- ✓ Graded Quiz: Introduction to LangChain in GenAI applications
  Graded Assignment • Grade: 100%

Module 3
Build a Generative AI Application with LangChain

Application Development Workflow with Generative AI

- ✓ Choose the Right AI Model for Your Use Case
  Video • 4 min
- ✓ From Idea to AI: Building Applications with Generative AI
  Video • 7 min
- ✓ Introduction to Flask
  Video • 7 min
- ✓ (Optional) Reading: Flask: A Gateway to Web Development in Python
  Reading • 15 min
- ○ **(Optional) Reading: Python with Flask for Large Scale Projects**
  Reading • 15 min
- ○ Cheat Sheet: Web Development Using Flask
  Reading • 5 min
- ○ Hands-on with GenAI: Choosing the Right Model for Your Application
  Ungraded App Item • 1h
- ○ Practice Quiz: Application Development Workflow with Generative AI
  Practice Assignment • 10 min

Summary and Evaluation

- ○ Summary and Highlights: Build a Generative AI Application with LangChain
  Reading • 3 min
- ○ Cheat Sheet: Build GenAI Application with LangChain
  Reading • 30 min
- ○ Graded Quiz: Build a Generative AI Application with LangChain
  Graded Assignment • 21 min

Course Wrap Up

- ○ Course Wrap-up
  Video • 4 min
- ○ Congratulations and Next Steps
  Reading • 2 min
- ○ Team and Acknowledgments
  Reading • 2 min

(Optional) Reading: Python with Flask for Large Scale Projects

0:05 / 4:04 1x

Introduction

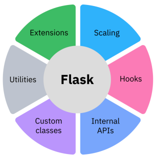
Python with Flask is a lightweight and flexible web application framework. It's known for its simplicity, minimalism, and ease of use. It is designed as a micro-framework providing a lightweight structure which facilitates developers in building a web application quickly and easily but not compromising on the efficiency and ability to scale up from small-scale projects to larger, more complex applications.

Python with Flask for large-scale development

Flask is a good choice for smaller, simpler applications. However, 'micro' has more to do with what Flask is, rather than limiting its scalability potential. Flask can be used for large-scale systems and more complex applications with attention to specific requirements and constraints, careful planning, good architecture, and modular design, but it may require more effort to manage and scale compared to more robust and feature-rich frameworks.

Its rich and robust ecosystem provides developers with tools, libraries, and functionalities to handle web development tasks such as routing, request handling, template rendering or similar tasks. Caching, load-balancing, replication and storing your data in a scalable manner can help achieve optimal results.

When building a large-scale application using Flask or when growing your codebase or scaling your application, following techniques can be considered:



Key Flask capabilities

- **Extensibility and integration:** Flask is extensible and developers can add or remove features enabling customization. Flask seamlessly integrates with other Python libraries and frameworks, enabling developers to combine its functionalities with other tools and technologies, thus enhancing its capabilities.
- **Transparent documentation:** Flask's documentation is published, allowing developers to use its internal APIs and utilities and find hook points, overrides and signals as per requirement.
- **Custom implementation:** Out of the box customizations and custom classes can be used for things like the request and response objects. The Flask class has many methods designed for subclassing. You can quickly add or customize behavior by subclassing Flask and using that subclass wherever you instantiate an application class.
- **Scaling considerations:** You can use scaling in such a way that if you double the number of servers you get about twice the performance. There is only one limiting factor regarding scaling in Flask which is the use of context local proxies. They depend on context which in Flask is defined as being either a thread, process or greenlet. If your server uses some kind of concurrency that is not based on threads or greenlets, Flask will no longer be able to support these global proxies.
- **Modular development:** Look for ways in which your project can be refactored into a collection of utilities and Flask extensions. Explore the many extensions in the community and look for patterns to build your own extensions if you do not find the tools you need. The best way to improve the tools for larger applications is getting feedback from users.

Real-world applications

Today, Python with Flask has become a popular choice among big names for its simplicity, flexibility, versatility and ease of learning and use. Its minimalistic design and customizable nature make it adaptable, effective, and reliable for large-scale web development requirements in diverse industries and sectors.

Several prominent companies including Netflix, Reddit, Lyft, LinkedIn, Pinterest and Uber leverage Python with Flask in their technology stacks for specific backend services or functionalities. Python Flask benefits big companies for diverse purposes such as API development, backend services, rapid development, and prototyping, while its extensibility facilitates the addition of functionalities within their infrastructure. This suggests that it can be part of scalable architectures when combined with appropriate strategies and tools.

Author(s)

Namrah Arif

Mark as completed

Like Dislike Report an issue

Go to next item →